1 CT reconstruction

(a) Set up the inverse problem. Since $\mu = w_k$ along the path length X_{kj} in pixel k,

$$y_j = -\ln \frac{I_{1,j}}{I_0} = \int_{\text{ray } j} \mu(l)\,\mathrm{d}l = \sum_{k=1}^{p} X_{kj} w_k, \qquad \boxed{\mathbf{y} = X^\mathsf{T}\mathbf{w}}, \quad \mathbf{y} \in \mathbb{R}^n,\ X \in \mathbb{R}^{p\times n},\ \mathbf{w} \in \mathbb{R}^p.$$

Known: X (scan geometry) and $\mathbf{y}$ (measurements). Unknown: $\mathbf{w}$. If $p > n$, then $\dim\ker(X^\mathsf{T}) \geq p-n$, and $\mathbf{w}_0 + \mathbf{v}$ fits the same data as $\mathbf{w}_0$ for every $\mathbf{v} \in \ker(X^\mathsf{T})$.

(b) Interpret the design matrix. Column j: path lengths of ray j through all pixels. In the image layout (99×117) it is a thin straight line. Row k: path lengths of all rays through pixel k. In the angle–detector layout (30×155) it is a sinusoidal curve, with one or two rays per angle.

(c) Design tests before using an agent. One justified test is enough. Two examples:

(i) For

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 3 \\ 8 \end{pmatrix}: \qquad \widehat{\mathbf{w}}_\lambda = \left(\frac{3}{1+\lambda}, \frac{16}{4+\lambda}, 0\right)^\mathsf{T} \xrightarrow{\lambda\to 0} (3, 4, 0)^\mathsf{T}.$$

The unmeasured w_3 stays zero for every λ. Reveals a transposed operator, a wrong penalty normalization or an added intercept.

(ii) Every angle sees the whole object. Rays are one pixel width apart, so the sum of y_j over the 155 rays of one angle approximates $\int \mu\,\mathrm{d}A$, independently of the angle. Hence the 30 row sums of the sinogram must agree (here 1649 ± 3), and $\sum_k \widehat{w}_k$ must equal them for small and moderate λ (within 0.3% for $\lambda \leq 10$). Reveals a centred or rescaled $\mathbf{y}$ or X, a fitted intercept and a wrong sinogram layout; the loss of total attenuation at large λ (-21% at 10^3) quantifies oversmoothing.

(d) Reconstruct the scan.

Checks: the tests in (c) pass; at every λ, $\|X(X^\mathsf{T}\widehat{\mathbf{w}}_\lambda - \mathbf{y}) + \lambda\widehat{\mathbf{w}}_\lambda\|_2 / \|X\mathbf{y}\|_2 < 10^{-11}$; a solver based on the eigendecomposition of $X^\mathsf{T}X$ agrees to $4 \cdot 10^{-8}$.

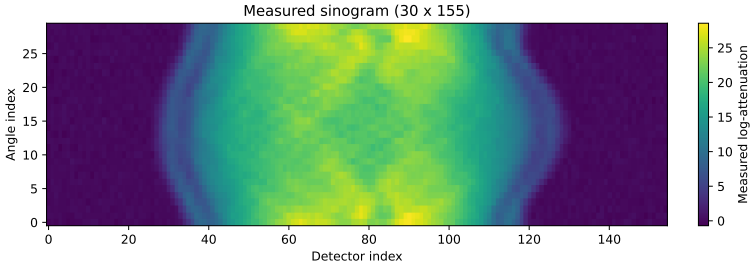


Figure 1: The supplied noisy measurements.

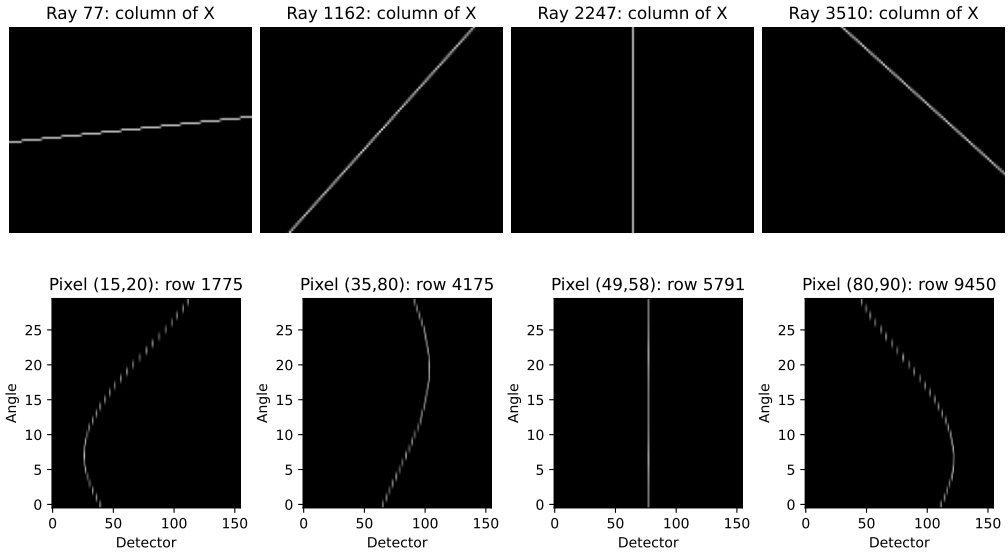


Figure 2: Selected columns (top) and rows (bottom) of X. Indices are zero-based.

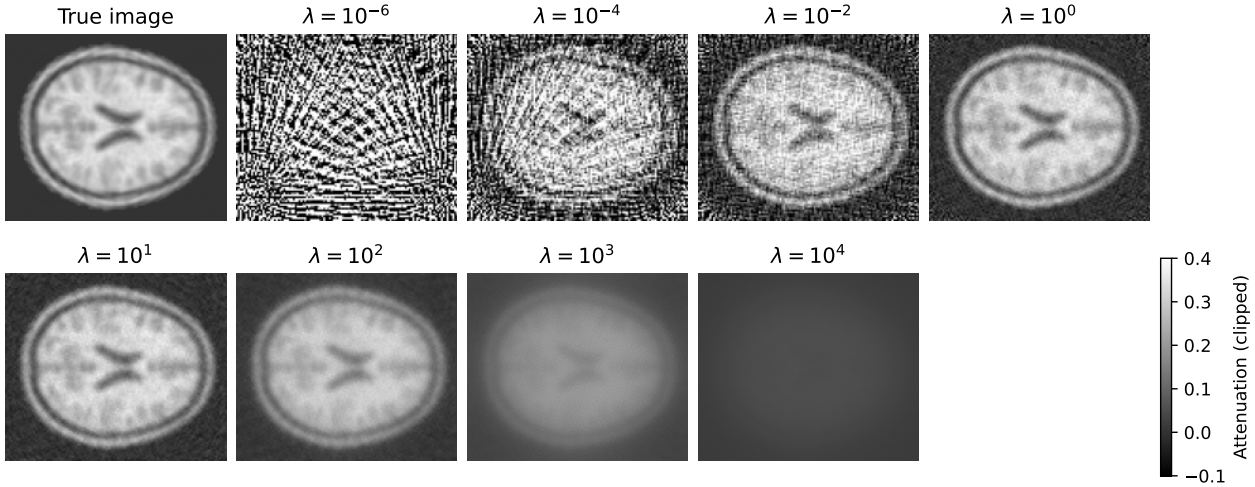


Figure 3: Simulated true image (not given to students) and ridge reconstructions, on one scale clipped to $[-0.1, 0.4]$.

λ	Data MSE	$\|\widehat{\mathbf{w}}_\lambda\|_2$	Minimum pixel	Maximum pixel	Image RMSE
10^{-6}	0.00547	137	-17.3	17.7	1.26
10^{-4}	0.00553	40	-5.22	4.17	0.319
10^{-2}	0.0057	23	-1.02	0.895	0.0935
10^{0}	0.00784	20.8	-0.19	0.407	0.0253
10^{1}	0.0209	20.5	-0.0689	0.377	0.0165
10^{2}	0.617	19.1	-0.0201	0.334	0.0303
10^{3}	17.6	13.6	0.0238	0.217	0.0829
10^{4}	128	4.11	0.0146	0.0594	0.159

Image RMSE is measured against the true image, so it cannot be computed from the information given in the original exercise. It is only provided for reference.

(e) Assess the reconstructions. For $\lambda \leq 10^{-2}$ the image is dominated by amplified noise. At $\lambda = 1$ and 10 the anatomy is clear. From $\lambda = 10^2$ on the image is oversmoothed. Accept reasoned choices near these boundaries. The residual grows monotonically from $5.5 \cdot 10^{-3}$ to 128.

A smaller residual does not mean a better image. From $\lambda = 10^{-6}$ to 10^{-2} the residual barely changes, while the image RMSE falls from 1.3 to 0.09. The grid's best image, at $\lambda = 10$, has about four times the smallest residual, and the residual cannot see null-space components.

2 Bayes decisions: signal or background?

(a) Posterior probability.

$$P(D) = 0.95 \cdot 0.05 + 0.1 \cdot 0.95 = 0.1425, \qquad \boxed{P(S \mid D) = \frac{0.95 \cdot 0.05}{0.1425} = \frac{1}{3}}.$$

Conditional losses: $2/3$ for retaining, $1/3$ for discarding. Under 0–1 loss, **discard**.

(b) Changing the loss.

True event	Retain	Discard
Signal	0	c_{FN}
Background	c_{FP}	0

The conditional risk is the expected loss over the unknown class:

$$r(a \mid x) = \mathbb{E}[L(Y, a) \mid x] = L(S, a)\, q + L(B, a)\,(1 - q),$$
$$r(\text{retain} \mid x) = c_{\mathrm{FP}}(1 - q), \qquad r(\text{discard} \mid x) = c_{\mathrm{FN}} q.$$

Retain when

$$c_{\mathrm{FP}}(1 - q) < c_{\mathrm{FN}} q \quad \Longleftrightarrow \quad \boxed{q > \frac{c_{\mathrm{FP}}}{c_{\mathrm{FP}} + c_{\mathrm{FN}}}}.$$

At equality both actions are optimal; choose discard as a fixed tie rule. With costs 1 and 10 the threshold is $1/11 \simeq 0.09091$, and

$$r(\text{retain} \mid D) = \frac{2}{3}, \qquad r(\text{discard} \mid D) = \frac{10}{3}.$$

So **retain**, reversing (a). The posterior is unchanged; only the losses differ.

(c) Bayes risk. Joint probabilities:

	D	$\bar{D}$	
S	$0.95 \cdot 0.05 = 0.0475$	$0.05 \cdot 0.05 = 0.0025$	0.05
B	$0.1 \cdot 0.95 = 0.095$	$0.9 \cdot 0.95 = 0.855$	0.95
	0.1425	0.8575	1

$$P(S \mid \bar{D}) = \frac{0.0025}{0.8575} = \frac{1}{343} \simeq 0.0029,$$

below both thresholds $1/2$ and $1/11$, so **discard** $\bar{D}$ under both losses. The Bayes risk (cf. Eq. 6.13) is

$$R = \sum_{x \in \{D, \bar{D}\}} \min_a r(a \mid x)\, P(x).$$

- *0–1 loss:* the rule discards everything:

$$R = P(D) \cdot \tfrac{1}{3} + P(\bar{D}) \cdot \tfrac{1}{343} = 0.0475 + 0.0025 = 0.05.$$

- $c_{\mathrm{FP}} = 1$, $c_{\mathrm{FN}} = 10$*:* the rule retains D and discards $\bar{D}$:

$$R = 1 \cdot P(B, D) + 10 \cdot P(S, \bar{D}) = 0.095 + 0.025 = 0.12.$$

Always discarding has risk $c_{\mathrm{FN}} P(S)$; always retaining has risk $c_{\mathrm{FP}} P(B)$:

Loss	Optimal rule	Always discard	Always retain
0–1	0.05	0.05	0.95
$c_{\mathrm{FP}} = 1,\ c_{\mathrm{FN}} = 10$	0.12	0.5	0.95

The check helps only if the threshold lies between the two posteriors. Under 0–1 loss both lie below $1/2$, so it changes no decision, like the lecture's pion–kaon classifier using $|\mathbf{p}|$ alone. With threshold $1/11$ it lowers the risk from 0.5 to 0.12.